Trace-preserving channel constraints on phase contrast and purity

Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). Port Coquitlam, British Columbia, Canada · 7 November 2026.

In the supplied four-use probability model, does the mean click rate necessarily rise after a balanced coherent error of ±60° about the commanded reflection? General density states can give either sign. A two-qubit channel has an additional constraint: its normalized Choi state has reference marginal $I_4/4$. Here the sign for the full channel class at ±60° remains open. The following checks give a narrower bracket.

Offset and model	Result (percentage points)
±60°, one shared response, general states	Sharp interval ≈ [+6.49019, +14.08960].
±60°, offset-specific general states, common detector	Sharp interval ≈ [−0.18064, +16.45926].
±60°, offset-specific channels, common detector	Minimum strictly exceeds the exact density-state lower endpoint. A feasible unitary pair gives +0.11820700… .
±61°, same channel class	The same unitary pair gives −0.02364942688… . A decrease is possible.

Fix P as a maximally entangled unitary match, h=0.99 as its detector click probability, s=131/320 as its population and q=81711338569043/224995905216000 as the ordinary click probability in each offset group. The readout phase is $U(\varphi)=P+\exp(i\varphi)(I-P)$. The ordinary and phase arms must use the same group weights. All tabulated contrasts are model probabilities. No matched outcomes after history 111 were supplied.

First offset at which a uniform rise can fail



The left endpoint is excluded for this channel class. The right endpoint is an upper bound from an explicit unitary pair. Its displayed position is rounded. The diagram brackets a mathematical threshold. No calibration measurements enter it.

A separate channel constraint strengthens the product-readout purity bound from 13/276 to 91/1230. The new coefficient is locally sharp for trace-preserving channels at each interior orthogonal-outcome population f (p. 4).

Why the density-state endpoint is excluded

All statements about the channel minimum here require one common effect and a rank-one, maximally entangled match. The input and output dimensions of the channel are both four.

Write $a=sh$, $t=a-q$, $b_j=\mathrm{Tr}[E(I-P)\rho_j(I-P)]$ and $z_j=2\,\mathrm{Im}\,\mathrm{Tr}[EP\rho_j(I-P)]$. With balanced signs the mean contrast is $(1+\cos\delta)A-\sin\delta\,B$, where $A=t+(b_+ +b_-)/2>0$ and $B=(z_+ -z_-)/2$. The attainable set is compact. Its largest ratio B/A, denoted κ_{ch}, is positive by the example on p. 3. The first zero of the worst-case contrast is $\delta_{ch}=2\arctan(1/\kappa_{ch})$ for $0<\delta<\pi$.

Equality would force incompatible unitary channels

The two positivity disks bound each $|z_j|/(t+b_j)$ by $\kappa=z_0/(t+b_0)$, with $b_0=(1-s)(1-h)$, $z_0=\sqrt{[4ab_0-(t+b_0)^2]}$. Equality requires the common corner. At 60° the unique minimizing support point is also that corner. At the corner, $|\mathrm{Tr}[EP\rho_j(I-P)]|^2=s(1-s)h(1-h)$. Equality in the vector Cauchy–Schwarz bounds forces both states pure, with orthogonal component along the same direction fixed by E. Indeed the positive Schur-complement remainder has zero trace. The two corner signs give opposite phases on this direction.

Writing normalized Choi kets as $|M\rangle=(M\otimes I)|\Phi_4\rangle$, take the match ket to be $|I\rangle$. Both operators $\sqrt{s}\,I+\sqrt{(1-s)}\exp(\pm i\gamma)V$ would have to be unitary, with $\mathrm{Tr}\,V=0$. Subtracting their unitarity equations forces V Hermitian. Adding them gives $V^2-\lambda V-I=0$, with $\lambda=(t+b_0)/[(1-s)\sqrt{(h(1-h))}]$. Trace zero with j positive eigenvalues and $4-j$ negative ones forces $\lambda^2=(4-2j)^2/[j(4-j)]$ in $\{0,4/3\}$. The prescribed λ^2 is 0.667689209796… . Thus both corner conditions cannot hold. Compactness excludes equality both for the ±60° density minimum and for $\kappa_{ch}=\kappa$.

> The exact general-state sign boundary $\delta_D=2\arctan(1/\kappa)$ is strictly below δ_{ch}. Page 3 gives $\delta_{ch}\le\delta_w$ with $60.83257915540° < \delta_w < 60.83257915542°$. Neither exclusion alone decides the sign at 60°.

What the variance correction adds

For $X=\cos\varepsilon$, let $m(x)$ be the angle-specific density-state lower support. Its corner branch has $m''(x)=z_0/(1-x^2)^{3/2}$. On the smooth branch $m''(x)\ge 2\sqrt{(aq)}/8>z_0$. The first derivatives match at the switch. Jensen applied to $m(x)-z_0x^2/2$ proves $E\Delta\ge m(EX)+z_0\,\mathrm{Var}(X)/2$. Since $m''(0)=z_0$, the uniform curvature constant is optimal. Balanced ±60° has $\mathrm{Var}(X)=0$. The variance result gives no improvement for that particular pair.

For a constant absolute offset, the density-state interval can also use pooled s and q if h is common and the two arms have matched group weights. Conjugate the interference coefficient in groups with negative offset to align the sine terms. Weighted Cauchy–Schwarz then gives both disk bounds on the pooled coefficients. The channel exclusion above uses s,h,q fixed in each group.

A certified unitary pair at 61°

These two diagonal unitaries are feasibility witnesses in the channel optimization. The construction gives an upper bound on the minimum; it makes no global optimality claim.

Let $z_4=(112+15i)/113$, $H=4\sqrt{s}-z_4$ and $m=(|H|^2-3)/(2|H|)$. Set $z_1=(H/|H|)(m+i\sqrt{(1-m^2)})$, $z_2=(H-z_1)/2$. The four diagonal entries of U are z_1,z_2,z_2,z_4. They all have modulus one: $|H-z_1|^2=4$ by construction. Their sum is $4\sqrt{s}$, so the unitary Choi state has match population s. Assign U to the negative offset and its complex conjugate to the positive offset.

Within the diagonal Choi subspace, p has coordinate vector $(1,1,1,1)/2$. Define orthonormal real vectors $v_1=(2,-1,-1,0)/\sqrt{6}$ and $v_2=(-1,-1,-1,3)/\sqrt{12}$. Put $u(y)=(v_1+y\,v_2)/\sqrt{(1+y^2)}$, $e=\sqrt{h}\,p+\sqrt{(1-h)}u(y)$, $E=|e\rangle\langle e|$. The effect is the same in both groups. Write $\alpha(y)=\langle u(y)|U\rangle$, using the normalized Choi vector $(z_1,z_2,z_2,z_4)/2$.

> The continuous scalar $|\sqrt{(sh)}+\sqrt{(1-h)}\alpha(y)|^2-q$ changes sign between $y=-0.014452126001853375$ and -0.014452126001853374. A root therefore sets the ordinary rate exactly in both groups. The verifier bounds the contrasts over this whole root enclosure.

Here $b=(1-h)|\alpha|^2$, $z=2\sqrt{[sh(1-h)]}\ \mathrm{Im}\ \alpha>0$. The gain in each group is $(1+\cos\delta)(t+b)-\sin\delta\, z$. Outward interval evaluation gives the strict brackets below, in percentage points.

Offset	Lower endpoint	Upper endpoint
60°	+0.11820700	+0.11820701
61°	−0.02364943	−0.02364942

The gain changes sign at $2\arctan[(t+b)/z]$. The interval checks at the two ends of the bracket on p. 2 have opposite signs. The diagonal-unitary form was found by exploring a restricted numerical family. The exact construction and interval checks establish feasibility independently of that search. The optimization over all channel states and effects remains unsolved.

A postselected operation need not have the input marginal $I_4/4$. The channel restriction applies to an apparatus only if that marginal, the same effective detector and the history-conditioned transfer assumptions can be established independently.

Kliesch and Roth, Theory of Quantum System Certification, PRX Quantum 2, 010201 (2021). Source ↗

White et al., Non-Markovian Quantum Process Tomography, PRX Quantum 3, 020344 (2022). Source ↗

A locally sharp product-readout purity bound

Separate fixed-detector problem. The complete specified phase curve is required. The ordinary click rate alone is insufficient.

Use $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$. Put $k=(sh-\sqrt{[shq]})^2/[sh(1-h)]$, $d=s+k$, $r=1-d$, $\eta=(\sqrt{s}\,p-\sqrt{k}\,u)/\sqrt{d}$ and $\theta=(\sqrt{k}\,p+\sqrt{s}\,u)/\sqrt{d}$. The fixed detector projects onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. The phase curve fixes the η population to d and forces θ into the kernel. Write $F=|v\rangle\langle v|$, $D=I-|p\rangle\langle p|-|u\rangle\langle u|-F$ and $f=\mathrm{Tr}(F\rho)$. Its remaining thirteen directions have total weight $r-f$.

The source gives $\rho_\star=d|\eta\rangle\langle\eta|+fF+(r-f)D/13$, $J(f)=\mathrm{Tr}(\rho_\star^2)$, and $p_0(f)=d+6(r-f)/13$ for the first-qubit Z-positive probability on input $|00\rangle$. The three fixed populations imply $\mathrm{Tr}(\rho\rho_\star)=J(f)$, hence $\mathrm{Tr}(\rho^2)=J(f)+\|\rho-\rho_\star\|_2^2$, where $\|\cdot\|_2$ is the Hilbert–Schmidt norm. All purities through one are possible on the supplied line $pZ=p_0(f)$.

> For a trace-preserving channel that transfers to the product input, $\mathrm{Tr}(\rho^2) \geq J(f)+(91/1230)[pZ-p_0(f)]^2$. At every interior $0<f<r$ the coefficient is attained for all sufficiently small deviations in either direction.

Proof. The probability observable is $M=4(|0\rangle\langle 0|\otimes I)_{\mathrm{out}} \otimes |00\rangle\langle 00|_{\mathrm{ref}}$. Let $B=I-|\theta\rangle\langle\theta|$. The population-only projection $M_1=BMB-|\eta\rangle\langle\eta|-6D/13$ has $\|M_1\|_2^2=276/13$. This gives the valid 13/276 coefficient in the supplied study. Its output partial trace is $L=6(|00\rangle\langle 00|-I/4)$: in BMB the marginal is $6|00\rangle\langle 00|+I/4$, and the subtractions remove $7I/4$. Thus $\mathrm{Tr}\,L^2=27$.

For traceless Hermitian H put $A(H)=B(I\otimes H)B$. It is orthogonal to the three population operators: η, v and θ are maximally entangled, and D is the remaining projection. Expanding B at θ gives $\mathrm{Tr}[A(H)A(G)]=(4-2/4)\mathrm{Tr}(HG)$ for traceless H,G. The trace-preserving direction is $M_2=M_1-(2/7)A(L)$. It has zero output partial trace and squared norm $\nu=276/13-(2/7)27=1230/91$. Thus $pZ-p_0=\mathrm{Tr}[M_2(\rho-\rho_\star)]$, and Cauchy–Schwarz gives the bound.

For sharpness set $\rho=\rho_\star+[(pZ-p_0)/\nu]M_2$. On the support B, $\rho_\star$ is strictly positive for interior f, so both signs of a sufficiently small perturbation preserve positivity. All constrained populations and the Choi marginal are unchanged. More explicitly, $|pZ-p_0|\leq\sqrt{\nu}\,\min\{d,f,(r-f)/13\}$ suffices. Equality is asserted in the trace-preserving class. No unital or mixed-unitary restriction is needed here.

The accompanying calculations check the channel constraints, the angular inequalities and the local purity coefficient. These are deterministic mathematical checks of model populations, not experimental observations.

K. Justus, Angle Moments and Variance, accompanying mathematical notes.

K. Justus, Purity After an Additional Readout, accompanying mathematical notes.